A DIMENSION FUNCTION ON A UNIVERSALLY CATENARY LOCALLY NOETHERIAN BASE PULLS BACK TO A DIMENSION FUNCTION ON ANY LOCALLY FINITE TYPE SCHEME OVER IT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-dimension-function-propagates is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 13656-13668 and proof span 13670-13738 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains a three-line display computing the difference of the two values of the candidate dimension function; the display and the author's reference to the dimension formula are preserved unchanged.

Editorial note. The author's notation for the candidate dimension function as both a function and a family indexed by the morphism is kept as written.

Editorial note. 6 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The same-chapter criterion for universal catenarity to be local, and the characterization of locally finite type morphisms by finite type ring maps on affine charts. Cited at proof lines 13677-13681. Makes the coordinate ring of an affine chart universally catenary and the corresponding ring map of finite type.
- The dimension formula for a finite type map of domains, cited from the algebra chapter as an equation, together with the description of points specializing to a given point through the spectrum of the local ring. Cited at proof lines 13719-13722 and 13689-13692. Supplies the identity relating transcendence degrees and dimensions of local rings used in the computation, and the reading of immediate specializations via the local ring's dimension.
- For a dimension function on a catenary scheme the value on a generic point minus the value on a specialization equals the codimension of the closure, and that codimension is the dimension of the local ring at the specialization.

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Cited at proof lines 13724-13728. Turns the difference of the values of the dimension function on the base into the dimension of the local ring of the base at the point.

Source reference. Original equation 02JV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3350–3350. Same-chapter reference used in the retained author text at source line(s) 13725. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Same-chapter reference used in the retained author text at source line(s) 2717, 2739, 2801, 2863, 3019, 3035, 3157, 3348, 3613, 3707, 3868, 5197, 5282, 6702, 9325, 9390, 9440, 11421, 13422, 13686. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02J9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3328–3328. Same-chapter reference used in the retained author text at source line(s) 3185, 3189, 3223, 3452, 3457, 3484, 13684. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02IZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3318–3318. Same-chapter reference used in the retained author text at source line(s) 13732. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01J7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2004–2004. Same-chapter reference used in the retained author text at source line(s) 1446, 12802, 13631, 13705, 13767. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02IA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3293–3293. Same-chapter reference used in the retained author text at source line(s) 13729. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a locally Noetherian and universally catenary scheme. Let $\delta : S \to \mathbf{Z}$ be a dimension function. Let $f : X \to S$ be a morphism of schemes. Assume f locally of finite type. Then the map*

$$\begin{aligned} \delta = \delta_{X/S} : X &\longrightarrow \mathbf{Z} \\ x &\longmapsto \delta(f(x)) + \mathit{trdeg}_{\kappa(f(x))}\kappa(x) \end{aligned}$$

is a dimension function on X.

Proof. Let $f : X \to S$ be locally of finite type. Let $x \rightsquigarrow y$, $x \neq y$ be a specialization in X. We have to show that $\delta_{X/S}(x) > \delta_{X/S}(y)$ and that $\delta_{X/S}(x) = \delta_{X/S}(y) + 1$ if y is an immediate specialization of x.

Choose an affine open $V \subset S$ containing the image of y and choose an affine open $U \subset X$ mapping into V and containing y. We may clearly replace X by U and S by V. Thus we may assume that $X = \mathrm{Spec}(A)$ and $S = \mathrm{Spec}(R)$ and that f is given by a ring map $R \to A$. The ring R is universally catenary (Lemma 02J9) and the map $R \to A$ is of finite type (Lemma 01T2).

Let $\mathfrak{q} \subset A$ be the prime ideal corresponding to the point x and let $\mathfrak{p} \subset R$ be the prime ideal corresponding to $f(x)$. The restriction δ' of δ to $S' = \mathrm{Spec}(R/\mathfrak{p}) \subset S$ is a dimension function. The ring $R/\mathfrak{p}$ is universally catenary. The restriction of $\delta_{X/S}$ to $X' = \mathrm{Spec}(A/\mathfrak{q})$ is clearly equal to the function $\delta_{X'/S'}$ constructed using the dimension function δ'. Hence we may assume in addition to the above that $R \subset A$ are domains, in other words that X and S are integral schemes, and that x is the generic point of X and $f(x)$ is the generic point of S.

Note that $\mathcal{O}_{X,x} = R(X)$ and that since $x \rightsquigarrow y$, $x \neq y$, the spectrum of $\mathcal{O}_{X,y}$ has at least two points (Schemes, Lemma 01J7) hence $\dim(\mathcal{O}_{X,y}) > 0$. If y is an immediate specialization of x, then $\mathrm{Spec}(\mathcal{O}_{X,y}) = \{x, y\}$ and $\dim(\mathcal{O}_{X,y}) = 1$.

Write $s = f(x)$ and $t = f(y)$. We compute

$$\begin{aligned}
\delta_{X/S}(x) - \delta_{X/S}(y) &= \delta(s) + \mathrm{trdeg}_{\kappa(s)}\kappa(x) - \delta(t) - \mathrm{trdeg}_{\kappa(t)}\kappa(y) \\
&= \delta(s) - \delta(t) + \mathrm{trdeg}_{R(S)}R(X) - \mathrm{trdeg}_{\kappa(t)}\kappa(y) \\
&= \delta(s) - \delta(t) + \dim(\mathcal{O}_{X,y}) - \dim(\mathcal{O}_{S,t})
\end{aligned}$$

where we use equality in (02JV) in the last step. Since δ is a dimension function on the scheme S and $s \in S$ is the generic point, the difference $\delta(s) - \delta(t)$ is equal to $\mathrm{codim}(\overline{\{t\}}, S)$ by Topology, Lemma 02IA. This is equal to $\dim(\mathcal{O}_{S,t})$ by Properties, Lemma 02IZ. Hence we conclude that

$$\delta_{X/S}(x) - \delta_{X/S}(y) = \dim(\mathcal{O}_{X,y})$$

and the lemma follows from what we said above about $\dim(\mathcal{O}_{X,y})$. □